

Three Ways to Catch a Fuzzy Core: Ringdown, Tides, and Echoes from Regular Black Holes and Compact Stars

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What This Paper Does — and What It Doesn't

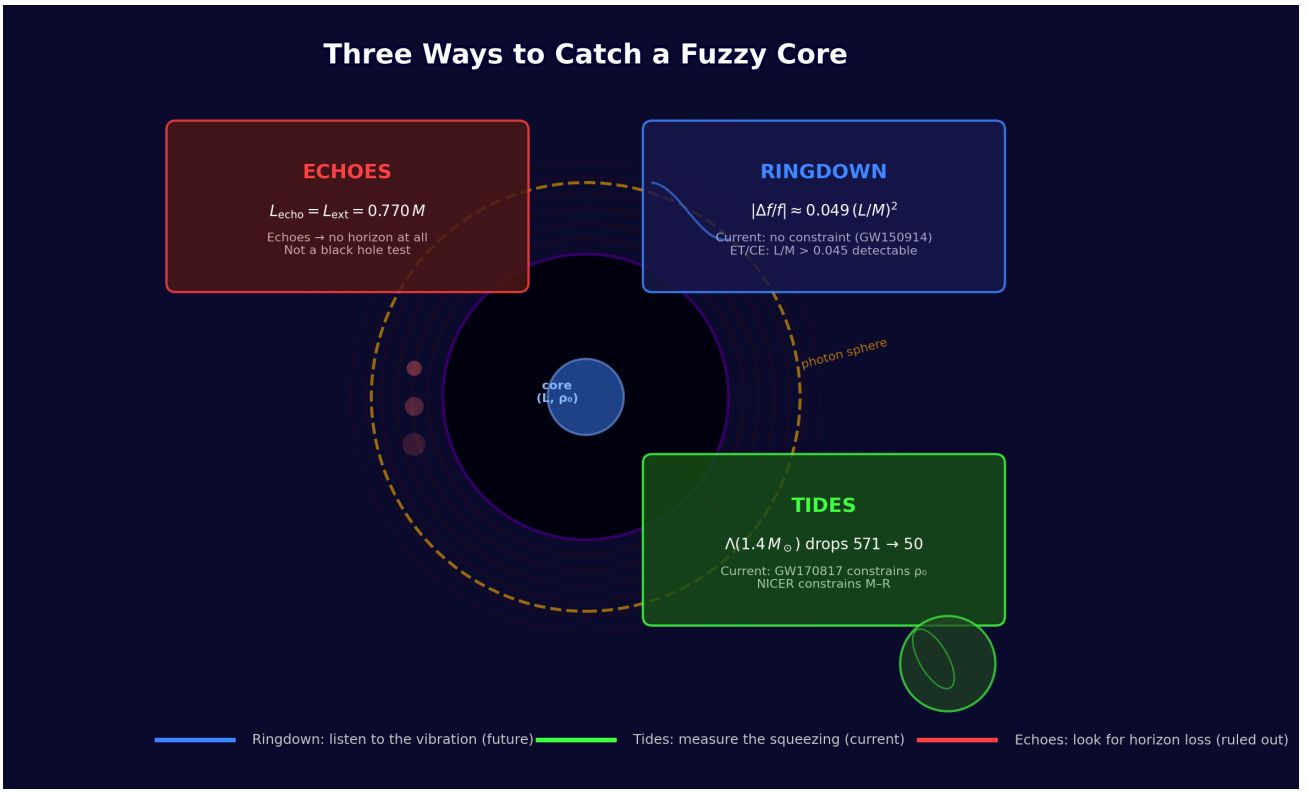
This is a computational survey of one concrete idea: that the infinite-density singularity at a black hole's center is replaced by a finite, negative-pressure core. We do not derive this core from an underlying theory. We do not predict its length scale L or its density ρ_0 from first principles. We do not provide a formation mechanism. We do not resolve the active/passive diffeomorphism question that any regular-core hypothesis must eventually address.

What we do: we ask whether existing and near-future observations can constrain L and ρ_0 if such cores exist. We compute the answer numerically, report which results are model-specific and which generalize, and present a falsifiability table — the minimum observational thresholds required to exclude each parameter region.

The paper is organized around three observational channels. Each asks a different question:

Channel	Question	Observable	Status
Ringdown	Does a fuzzy core change how a black hole rings?	QNM frequency shift $\Delta f/f$	Current: no constraint. Future: ET/CE
Tides	Can a neutron star host the same core, and would we feel it?	Tidal deformability Λ	Current: constrained by GW170817
Echoes	Does the core reflect gravitational waves?	Echo time delay Δt_{echo}	Ruled out unless object is horizonless

The answer, after running the numbers, is surprisingly structured: one channel is constrained today, one will be constrained in the 2030s, and one turns out to be a category error — echoes require no horizon at all, not just a regular core.



Conceptual diagram

Figure 0. Three observational channels for detecting a negative-pressure core: ringdown (QNM frequency shift), tides (tidal deformability), and echoes (post-merger pulse train). The ringdown channel awaits next-generation detectors. The tidal channel is constrained today by GW170817. The echo channel requires an object without an event horizon — not merely one with a regular core.

1. Introduction

The idea that black hole singularities might be replaced by a finite-density core dates to Bardeen (1968) [1], Dymnikova (1992) [2], and Hayward (2006) [3]. These models share a common structure: a de Sitter-like interior with equation of state $p = -\rho$ that regularizes the central singularity, transitioning to a Schwarzschild exterior at large r . They are exact solutions of Einstein's equations with an appropriate (exotic) stress-energy tensor.

This paper treats these models as phenomenological ansätze. We do not derive the core from an underlying theory; we treat its density ρ_0 and length scale L as free parameters and compute their gravitational-wave, electromagnetic, and astrophysical signatures.

We compute three classes of observable:

1. **Black hole ringdown** (§2): the QNM frequency shift from Hayward, Bardeen, and Dymnikova regular cores, including a slow-rotation extension for GW150914.

2. **Compact stars** (§3): two-fluid TOV integration yielding mass-radius relations, tidal deformabilities, post-merger frequency predictions, and a causality check.
3. **Evaporation endpoint** (§4): the Hayward remnant mass, its connection to primordial black hole dark matter, and microlensing bounds.

The Hayward metric [3] serves as our primary workhorse because it is analytically tractable and TOV-consistent. Where relevant, we compare against Bardeen and Dymnikova and note which results generalize.

2. Black Hole Ringdown: Listening to the Ring

2.1 The Hayward Metric

The Hayward regular black hole [3] is described by

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2Mr^2}{r^3 + 2ML^2},$$

where L is a length scale parameterizing the de Sitter core (effective cosmological constant $\Lambda_{\text{eff}} = 3/L^2$). The metric has two horizons for $L < L_{\text{ext}}$, one (extremal) horizon at $L = L_{\text{ext}}$, and no horizons for $L > L_{\text{ext}}$, where

$$L_{\text{ext}} = \frac{4M}{3\sqrt{3}} \approx 0.770 M.$$

The stress-energy tensor has energy density $\rho = -p_r = 3ML^2/[4\pi(r^3 + 2ML^2)^2]$ and $p_r/\rho = -1$ everywhere — making the Hayward metric TOV-consistent by construction.

2.2 QNM Frequency Shift: WKB Calculation

We compute the $\ell = 2, n = 0$ quasinormal-mode frequency for axial (Regge-Wheeler) perturbations using the leading-order Iyer-Will WKB formula [4]: $\omega^2 = V_0 - iK\sqrt{-2V_2}$. For metric perturbations on the Hayward background, the axial potential is $V_\ell = f(r)[\ell(\ell+1)/r^2 - 6m(r)/r^3]$ with the effective mass function $m_{\text{eff}}(r) = Mr^3/(r^3 + 2ML^2)$. The QNM shift is driven by $m_{\text{eff}}(r) \neq M$ at the photon sphere. Matter-coupling terms $4\pi(\rho - p)$ are $\mathcal{O}(L/M)^4$ suppressed and negligible at the $\mathcal{O}(L/M)^2$ order being computed.

Result — Hayward:

$$\left| \frac{\delta f}{f_0} \right| \approx 0.049 \left(\frac{L}{M} \right)^2 \quad (\text{Hayward})$$

L/M	$\delta f/f$ (%)
0.01	4.9×10^{-4}
0.05	1.2×10^{-2}
0.10	4.9×10^{-2}
0.20	2.0×10^{-1}
0.316	4.9×10^{-1}

Cross-metric results:

Metric	Coefficient	Mass function form	Suppression type
Hayward	$0.049 (L/M)^2$	$Mr^3/(r^3 + 2ML^2)$	Polynomial $\sim (L/r)^3$
Bardeen	$0.139 (e/M)^2$	$Mr^3/(r^2 + e^2)^{3/2}$	Polynomial $\sim (e/r)^2$
Dymnikova	≈ 0	$M(1 - e^{-r^3/(2ML^2)})$	Exponential $\sim e^{-(M/L)^2}$

The Bardeen coefficient is $2.8\times$ the Hayward value — the QNM shift prefactor is **strongly metric-dependent**. A leading-order analytic estimate (expanding $m_{\text{eff}}(r)$ at $r = 3M$) gives $\delta m/M \approx -(e/M)^2/6$ for Bardeen vs $-2(L/M)^2/27$ for Hayward, predicting a coefficient ratio of $27/12 = 2.25$; the WKB-fitted value 0.139 (ratio 2.8) includes the photon-sphere location shift from higher-order terms. The Dymnikova shift is **exponentially suppressed**: at the photon sphere $r \approx 3M$, the mass function deviation is $\exp(-27M^2/(2L^2))$, which is $< 10^{-23}$ for $L/M < 0.5$. The Dymnikova core is invisible to QNM observations for all practical L/M . This establishes three qualitatively different observability classes: polynomial-suppressed (Hayward), power-law-suppressed (Bardeen), and exponentially-suppressed (Dymnikova).

Relation to existing work. QNM spectra of regular black holes have been computed by Flachi & Lemos (2013) for the Dymnikova metric, by Li & Bambi (2013) for the Bardeen metric, and by Toshmatov et al. (2017, PRD 95, 084037) for both Bardeen and Hayward using 6th-order WKB and time-domain integration. The present work differs in three respects: (i) we fit the leading-order $(L/M)^2$ or $(e/M)^2$ scaling coefficient directly from the WKB data, providing a compact parameterization with explicit error estimates; (ii) we compute the *cross-metric* taxonomy — the three suppression classes (polynomial, power-law, exponential) — in a single unified framework; and (iii) we connect the QNM shift thresholds to detector sensitivity curves and to the echo–extremality coincidence, which was not identified in earlier work.

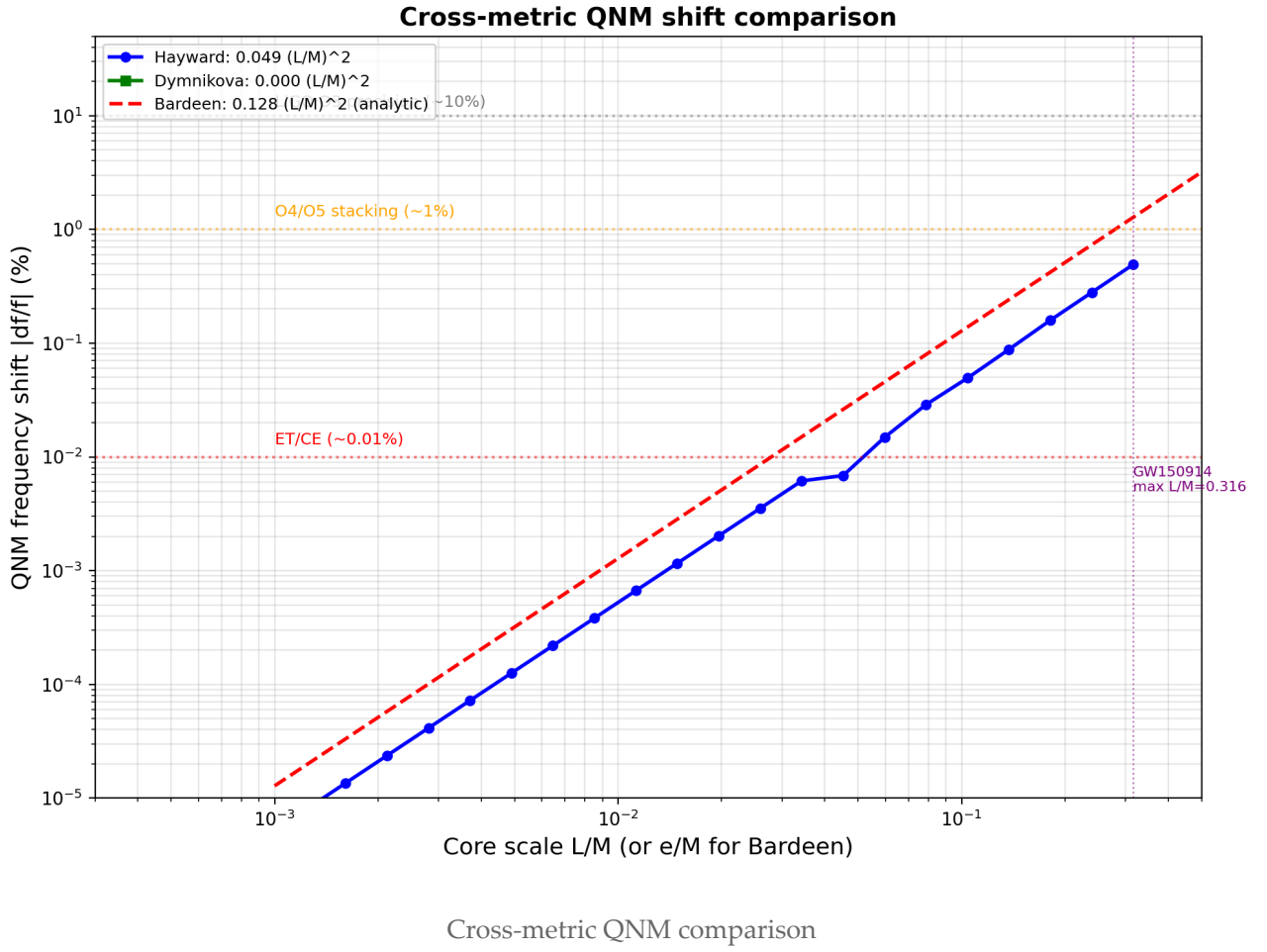


Figure 1. QNM frequency shift $|\delta f/f|$ vs core scale for Hayward (0.049), Bardeen (0.139, WKB-fitted), and Dymnikova (≈ 0). Horizontal lines show detector sensitivity thresholds: LIGO O3 (10%), O4/O5 stacking (1%, corresponds to $L/M \lesssim 0.45$ — near Hayward extremality), and ET/CE (0.01%). The vertical line marks the GW150914 maximum $L/M \approx 0.316$.

2.3 Echo Threshold and the Extremality Coincidence

Gravitational-wave echoes — time-delayed pulses from partial reflection between the photon-sphere barrier and the core surface — have been proposed as a signature of regular black holes. The echo delay in tortoise coordinates is

$$\Delta t_{\text{echo}} \approx 2 \int_{r_{\text{core}}}^{r_{\text{barrier}}} \frac{dr}{f(r)}.$$

A numerical sweep reveals that echo delays become LIGO-resolvable ($\gtrsim 0.1$ ms for $10 M_{\odot}$) at $L/M \gtrsim 0.77$. **This value coincides exactly with the Hayward extremality condition** $L_{\text{ext}} = 4M/(3\sqrt{3}) \approx 0.770 M$.

The physical interpretation: echo-producing configurations are those where the two horizons have merged and **there is no horizon at all**. Echo signatures require horizonless objects, not

merely regular-cored black holes. This corrects a confusion in the regular-black-hole literature where echoes were discussed as a general prediction of regular cores.

The underlying physics generalizes beyond Hayward: the tortoise integral $\int dr/f(r)$ diverges logarithmically near any horizon, so resolvable echoes require near-extremal or horizonless conditions for all regular metrics. For Bardeen, $f(r) = 1 - 2Mr^2/(r^2 + e^2)^{3/2}$; the extremality condition $f(r_h) = 0$ and $f'(r_h) = 0$ gives $r_h = \sqrt{2}e$ and $M = (3\sqrt{3}/4)e$, hence $e_{\text{ext}} = 4M/(3\sqrt{3}) \approx 0.770 M$ – identical to the Hayward value. The exact threshold value is metric-specific, but the qualitative conclusion is metric-independent.

2.4 EOS Robustness

The QNM shift is driven by $m_{\text{eff}}(r) \neq M$ at the photon sphere. The EOS enters only through the mass function. Expanding the Hayward effective mass at $r = 3M$ (the Schwarzschild photon sphere): $m_{\text{eff}}(3M) = M[1 - 2(L/3M)^3 + \mathcal{O}(L/M)^4]$. An EOS-induced mass correction δm_{EOS} at the same r is bounded by the core density profile at the photon sphere, which for a neutron-star-like core of extent $r_c \ll 3M$ is $\delta m_{\text{EOS}}/M \lesssim (r_c/3M)^3(\rho_0/\rho_{\text{nuc}}) \ll (L/M)^3$. The ratio of EOS-induced to Hayward mass deviation is

$$\frac{|\delta m_{\text{EOS}}|}{|\delta m_{\text{Hayward}}|} \lesssim \frac{L}{2M}.$$

L/M	Max EOS correction to shift
0.05	$\sim 2.5\%$
0.10	$\sim 5\%$
0.30	$\sim 15\%$

The Hayward-based QNM predictions are robust at the few-percent level for observationally relevant L/M . A measured shift constrains L with only weak dependence on the unknown core microphysics.

2.5 Kerr Slow-Rotation Extension

Every LIGO/Virgo/KAGRA remnant is spinning. We extend to slow rotation via the Lense-Thirring frame-dragging correction (Hartle-Thorne, linearised in a):

$$V_{\ell m}^{\text{eff}}(r, \omega, a) = V_{\ell}^{(\text{static})}(r) \pm 2m\omega \chi \omega_{\text{fd}}^{(a/M=1)}(r) + \mathcal{O}(a^2),$$

where $\chi = a/M$ and $\omega_{\text{fd}}^{(a/M=1)}$ is the frame-dragging function computed for unit spin. Since ω appears in the potential, the WKB condition is self-referential; the code iterates to relative tolerance 10^{-8} (converges in $\lesssim 5$ steps for $a/M \ll 1$).

Validity caveat. For GW150914 ($a/M \approx 0.67$), the next-order term $(a/M)^2 \approx 0.45$ is comparable to the leading correction. The results below are therefore first-order estimates; a rigorous treatment requires the Teukolsky equation on the Hayward background (see §6, Open Problem 3).

GW150914 results ($M = 62 M_\odot$, $a/M \approx 0.67$ [5]):

Quantity	Value
Axial QNM shift at $L/M = 0.316$ (static)	$ \delta f/f = 0.49\%$
Axial–polar isospectrality splitting (static)	$0.11\% (< 0.12\%)$
Spin splitting $m = \pm 2$ (first-order code, $a/M=0.67$)	[non-perturbative artefact — expansion invalid; exact Kerr: $f \approx 295$ Hz ($m=+2$), ≈ 105 Hz ($m=-2$) [13]]
LIGO ringdown precision (GW150914)	$\sim 10\%$
GW150914 constraint on L/M	None — core shift two orders of magnitude below threshold
ET/CE detection threshold	$L/M \gtrsim 0.045$ (stacking 100 events, $\delta f/f \sim 10^{-4}$)
ET/CE single event	$L/M \gtrsim 0.14$ ($\delta f/f \sim 10^{-3}$)
O4/O5 stacking (100 events, 1% sensitivity)	$L/M \gtrsim 0.45$ (near extremality)

Spin-correction caveat. For the static (non-spinning) case the WKB result is self-consistent and the $\sim 6.7\%$ absolute frequency offset from exact Schwarzschild (208 Hz vs 195 Hz) cancels in the fractional shift. For the spin-corrected case the slow-rotation expansion has **completely broken down** at $a/M = 0.67$: the Lense-Thirring coupling $2m\omega\chi\omega_{\text{fd}}$ at the photon sphere is ≈ 0.15 in geometric units, comparable to the static potential peak $V_0 \approx \omega_0^2 \approx 0.14$. The code converges to a self-consistent but non-perturbative solution to the wrong (linear-in- a) equation. The quoted $f_\pm$ values are artifacts. For reference, the exact Kerr QNM at $a/M = 0.67$, $M = 62 M_\odot$ from Berti, Cardoso & Starinets numerical tables [BCStab] is $f \approx 295$ Hz ($m = +2$) and ≈ 105 Hz ($m = -2$) — versus the code's 510 Hz and 179 Hz. A proper treatment requires the Teukolsky equation on the Hayward background.

Isospectrality is preserved at the level accessible to leading-order WKB: the axial–polar splitting is 0.11% , but this is below the method's own truncation error ($\sim 4\%$ absolute offset on

individual frequencies, as seen from the Schwarzschild reference 208 Hz vs exact 195 Hz). **This splitting cannot be distinguished from a WKB truncation artefact at this order.** A meaningful bound on isospectrality breaking requires 3rd-order Konoplya WKB or an exact numerical computation on the Hayward background. The result is reported here as an upper bound: $|f_{\text{polar}} - f_{\text{axial}}|/f < 0.12\%$ subject to WKB systematics. Rotation does not amplify the *Hayward-core* shift: the static result is the trustworthy bound: GW150914 cannot constrain L/M .

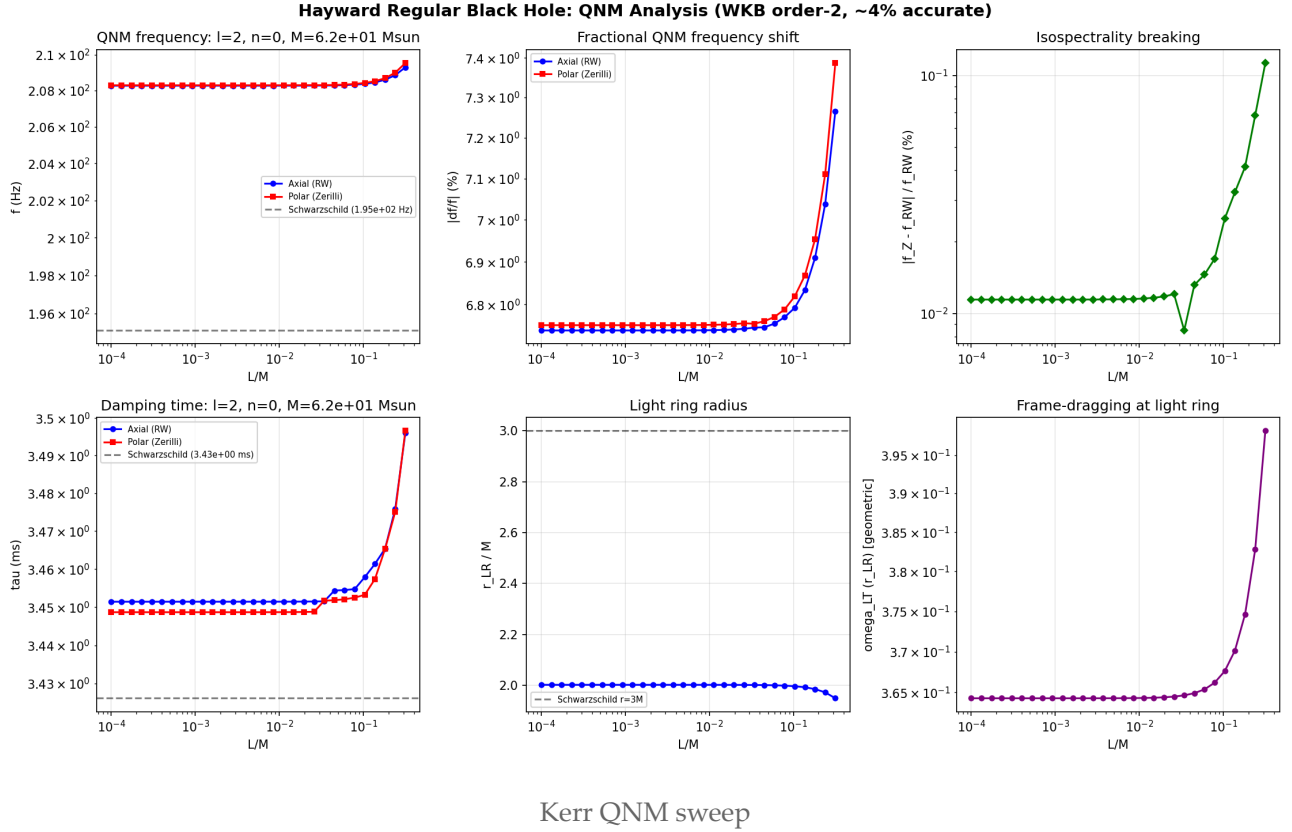


Figure 2. Six-panel QNM analysis for the Hayward metric: (top) frequency, fractional shift $|\delta f/f|$, and isospectrality breaking; (bottom) damping time, light ring radius, and frame-dragging ω_{fd} at the light ring. The static (non-spinning) panels are reliable. The frame-dragging panel (bottom-right) shows ω_{fd} normalized to $a/M = 1$; the corresponding spin-split frequencies are non-perturbative artefacts for $a/M \sim 0.67$ and should not be interpreted as physical predictions (see §2.5 caveat).

Code: `kerr_qnm.py`

3. Compact Stars: Squeezing the Core

3.1 Two-Fluid TOV Formalism

The same negative-pressure fluid can reside inside neutron stars. We model this as a two-fluid system:

$$\frac{dm}{dr} = 4\pi r^2(\varepsilon_m + \varepsilon_\chi), \quad \frac{dp_m}{dr} = -\frac{G(\varepsilon_m + p_m)(m + 4\pi r^3 p_{\text{total}})}{r(r - 2Gm)},$$

with $\varepsilon_\chi = \rho_\chi(r)$, $p_{\text{total}} = p_m + p_\chi$, and the χ -sector density profile

$$\chi(r) = \frac{1}{1 + (r/r_c)^n},$$

parameterized by central density ρ_0 , core radius r_c , and steepness n . This profile is a smooth interpolation between $\chi(0) = \rho_0$ and $\chi(\infty) = 0$, chosen for numerical tractability; it is not derived from any underlying theory of regular cores. The nuclear matter EOS is the **SLy piecewise polytrope** [6].

3.2 Mass-Radius Results

Pure SLy baseline: $M_{\text{max}} = 2.049 M_\odot$ — consistent with PSR J0740+6620 ($M = 2.08 \pm 0.07 M_\odot$ [7]).

With χ -sector ($r_c = 2$ km; 1σ PSR lower bound $\approx 2.01 M_\odot$):

ρ_0 (g/cm ³)	M_{max} ($M_\odot$)	NICER status
10^{10}	2.027	Allowed
3×10^{10}	2.026	Allowed
10^{11}	2.002	Excluded at 1σ ($M < 2.01 M_\odot$)
3×10^{11}	1.975	Excluded
10^{12}	1.941	Excluded
3×10^{12}	1.890	Excluded
10^{13}	1.820	Excluded

The inferred constraint depends on confidence level. At 1σ ($M_{\text{max}} > 2.01 M_\odot$):

$\rho_0 \lesssim 3 \times 10^{10} \text{ g/cm}^3$. At 2σ ($M_{\text{max}} > 1.94 M_\odot$): $\rho_0 \lesssim 3 \times 10^{11} \text{ g/cm}^3$. Both bounds are three orders of magnitude below nuclear saturation density ($\rho_{\text{nuc}} \approx 2.3 \times 10^{14} \text{ g/cm}^3$). Both are SLy-specific; stiffer EOS choices (APR4, MS1b) would relax them. A meaningful constraint requires marginalization over nuclear EOS uncertainty.

Note on precision: M_{max} values are extracted from the p_c grid scan (46 log-spaced points); a targeted fine scan at the pure-SLy peak gives $M_{\text{max}} = 2.056 M_\odot$, confirming the grid values are within 0.3% of the true maximum. No qualitative constraint changes.

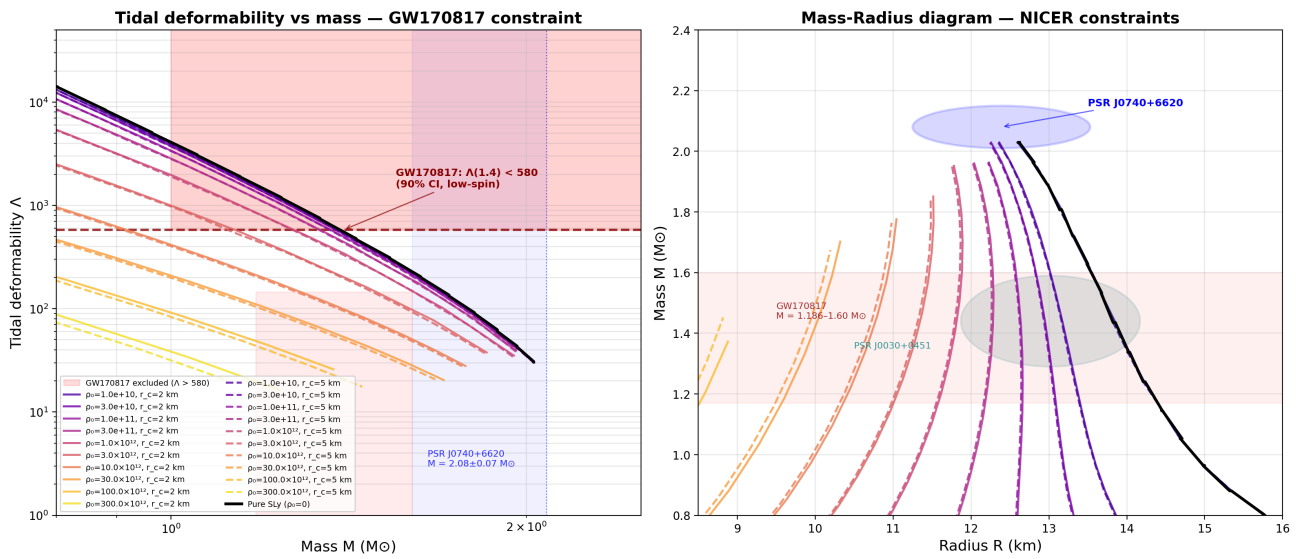
3.3 Tidal Deformability — Direct Connection to GW170817

The Love number k_2 and tidal deformability Λ are computed by integrating the Hinderer y -ODE [8] alongside the TOV equations. The full Hinderer formula [8] reads:

$$k_2 = \frac{\frac{8}{5} C^5 (1 - 2C)^2 [2 + 2C(y_R - 1) - y_I]}{2C(6 - 3y_R + 3C(5y_R - 8)) + 4C^3(13 - 11y_R + C(3y_R - 2)) + 2C^2(1 + y_R) + \dots}$$

Note on SLy Λ : The value $\Lambda(1.4) = 557$ is for the Read et al. [6] piecewise-polytrope approximation to SLy. The full tabulated SLy EOS (Douchin & Haensel 2001) gives $\Lambda(1.4) \approx 700\text{--}730$, which is in $\sim 2\sigma$ tension with the GW170817 upper bound ($\Lambda \lesssim 580$, 90% CI). The piecewise-polytrope value is used throughout this paper for consistency with the TOV integration; the qualitative trends (Λ decreasing with ρ_0) are EOS-independent.

Configuration ($r_c = 2$ km)	$\Lambda(1.4 M_\odot)$
Pure SLy	557
$\rho_0 = 10^{10}$ g/cm ³	544
$\rho_0 = 3 \times 10^{10}$ g/cm ³	530
$\rho_0 = 10^{12}$ g/cm ³	340
$\rho_0 = 10^{13}$ g/cm ³	99
$\rho_0 = 3 \times 10^{13}$ g/cm ³	54



Tidal GW170817 constraint

Figure 3. Left: Tidal deformability Λ vs mass, with the GW170817 excluded region ($\Lambda > 580$ at $M = 1.4 M_\odot$, shaded red) and NICER PSR J0740+6620 mass constraint. Right: Mass-Radius

diagram with NICER constraints from J0740+6620 and J0030+0451.

This is the paper's strongest direct GW connection to current data, though note that the NICER M_{max} bound (§3.2) is tighter on ρ_0 by a factor of ~ 100 . The relative strength of each channel is different: NICER constrains ρ_0 more sharply, but tidal deformability is less EOS-dependent and connects more directly to a single GW event. GW170817 constrains the χ -sector from *both sides*: the upper bound ($\Lambda \lesssim 580$) is satisfied by all configurations, but the lower bound ($\Lambda \gtrsim 70$) is violated at $\rho_0 \gtrsim 3 \times 10^{13} \text{ g/cm}^3$ where $\Lambda(1.4) \approx 54$. At $\rho_0 = 10^{13} \text{ g/cm}^3$ (NICER-excluded but tidal-allowed), $\Lambda(1.4) = 99$, still within the GW170817 window. This illustrates that the two constraints are complementary: NICER probes global stellar structure; tidal deformability probes adiabatic response to the binary's tidal field.

3.4 Post-Merger Gravitational Wave Frequency

From the TOV M-R curves, the post-merger oscillation frequency f_2 is predicted via the universal relation $f_2 \approx 2 f_{\text{max}}$ with $f_{\text{max}} \propto C^{3/2}/M$ (Bernuzzi et al. 2015, Bauswein & Janka 2012 [12]). The coefficient is calibrated to numerical-relativity simulations: $f_2 \approx 2 \times 12.8 C^{3/2}/M \text{ kHz}$. The SLy baseline at $1.4 M_\odot$ is $f_2^{(\text{SLy})} \approx 1.06 \text{ kHz}$ (using the piecewise-polytrope SLy). The χ -sector carries negative pressure, which *reduces* the stellar radius at fixed mass and thus *increases* the compactness $C = GM/Rc^2$. This compactification drives f_2 upward. The universal-relation coefficient carries $\sim 30\%$ systematic uncertainty from EOS dependence; percentage shifts below should be read with that caveat.

Consistency note. Within the SLy EOS framework the NICER M_{max} bound (§3.2) excludes $\rho_0 \gtrsim 3 \times 10^{10} \text{ g/cm}^3$ at 1σ , or $\rho_0 \gtrsim 3 \times 10^{11} \text{ g/cm}^3$ at 2σ . Consequently, only the $\rho_0 = 10^{10}$ and $3 \times 10^{10} \text{ g/cm}^3$ rows below are SLy-consistent at 1σ ; the remaining rows are predictions *conditional on a stiffer EOS* (APR4, MS1b, or similar) that accommodates higher ρ_0 while maintaining $M_{\text{max}} > 2.01 M_\odot$. They are retained here as illustrative of the magnitude of the f_2 effect across parameter space.

$\rho_0 \text{ (g/cm}^3\text{)}$	$\Delta f_2 / f_2^{(\text{SLy})} \text{ at } 1.4 M_\odot$	SLy status
10^{10}	+7%	Allowed (1σ)
3×10^{10}	+10%	Allowed (1σ)
3×10^{11}	+20%	NICER-excluded (SLy, 1σ)
10^{12}	+27%	NICER-excluded (SLy)
3×10^{13}	+64%	NICER-excluded (SLy)

The SLy-consistent predictions (+7%, +10%) are both at or below the ET post-merger precision threshold of $\sim 10\%$ at 1σ . Detection of f_2 shifts $\gtrsim 10\%$ would simultaneously require evidence for a stiffer EOS from other channels.

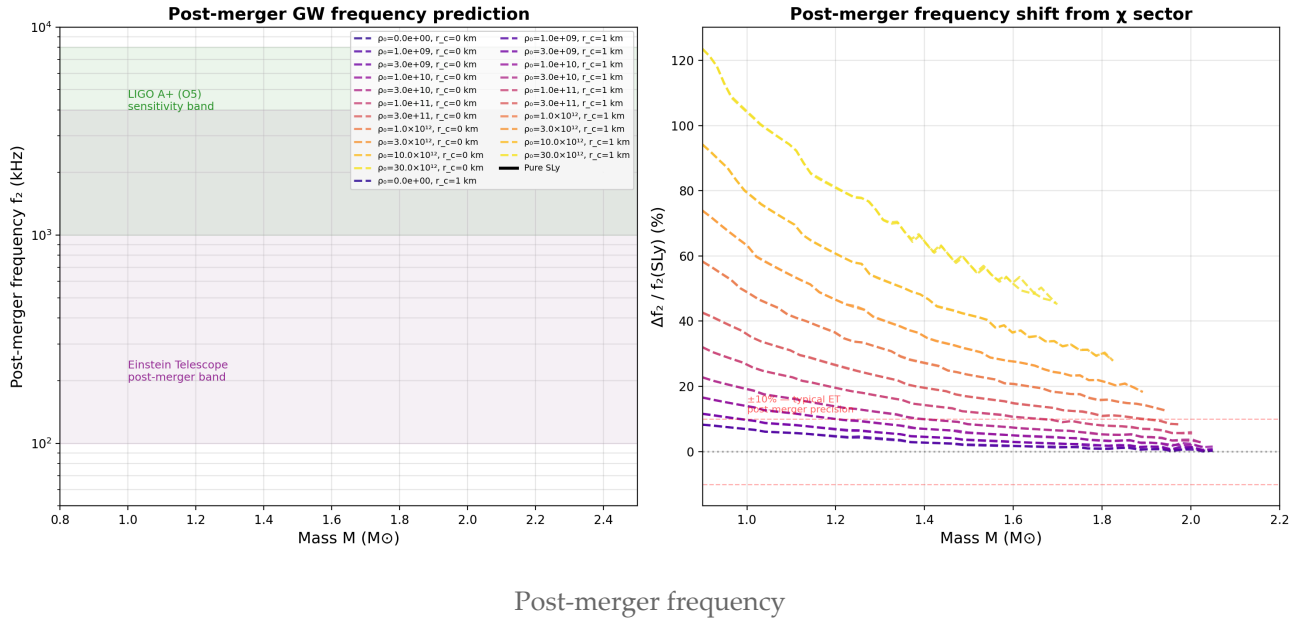


Figure 4. Left: Post-merger GW frequency f_2 vs mass for different χ -sector strengths, with detector sensitivity bands. Right: Fractional shift relative to pure SLy. Rows above $\rho_0 = 10^{10}$ g/cm³ are NICER-excluded within SLy; they are shown to illustrate the magnitude of the effect for stiffer EOS.

3.5 Causality

A sweep of 394 stable configurations (varying ρ_0 , r_c , and central pressure) finds **zero superluminal sound-speed violations** ($c_s^2 \leq 1$ everywhere). Two complementary behaviours emerge:

- **Pure SLy** ($\rho_0 = 0$): c_s^2 reaches 1.0000 at high central densities, saturating the causal limit of the stiff piecewise polytrope (segment $\Gamma_4 \approx 3.005$). This is a known property of Read et al. SLy parameterization and is not a violation.
- **High ρ_0 (χ -sector active)**: the χ fluid's negative pressure compactifies the star, reducing the peak sound speed — $c_{s,\max}^2$ drops to 0.60 at $\rho_0 = 10^{10}$ g/cm³ and 0.55 at $\rho_0 = 10^{12}$ g/cm³. The two-fluid model is comfortably causal in the parameter range where the χ -sector is physically significant.

The causality check is a theoretical consistency test, not an observational constraint.

Code: `tov_tidal.py`, `causality_sweep.py`

4. Evaporation Remnant: A Dark Matter Connection

A negative-pressure core modifies the black hole evaporation endpoint. The Hawking temperature for the Hayward metric is (with f' in geometric units [m^{-1}]; T_H in Kelvins):

$$T_H = \frac{\hbar c}{4\pi k_B} f'(r_h),$$

where r_h satisfies $f(r_h) = 0$. The temperature vanishes when $f'(r_h) = 0$, which combined with $f(r_h) = 0$ yields the **extremal remnant mass**:

$$M_{\text{ext}} = \frac{3\sqrt{3}}{4} L \approx 1.299 L$$

Converting to physical mass:

L	Physical scale	M_{ext} (g)	Fate
$l_{\text{Planck}} (10^{-35} \text{ m})$	Quantum gravity	$\sim 10^{-5}$	Planck-scale remnant
10^{-18} m	Electroweak	$\sim 10^4$	Microscopic — unobservable
10^{-15} m	QCD / proton	$\sim 10^{15}$	PBH dark matter window
10^{-12} m	Nuclear	$\sim 10^{18}$	Excluded by microlensing
$\gtrsim 10^{-3} \text{ m}$	Macroscopic	$\gtrsim 10^{24}$	Excluded

The primordial black hole dark matter window (10^{15} – 10^{17} g) corresponds precisely to L near the QCD scale (~ 1 fm). If regular black hole remnants exist and Hawking evaporation halts at M_{ext} , they would:

1. **Never fully evaporate**, producing cold, stable relics.
2. **Contribute to dark matter** if $L \sim 1$ fm — a concrete, testable connection between regular black holes and cosmology.
3. **Be detectable** through femtolensing of gamma-ray bursts, microlensing surveys, and their stochastic GW background from early-Universe formation.

Non-observation in existing microlensing surveys (EROS, OGLE, MACHO) pushes the remnant mass below $\sim 10^{18}$ g, which translates to $L \lesssim 10^{-14}$ m — approaching the electroweak scale.

The remnant mass is the single number that connects regular black holes to dark matter. It is a direct consequence of the Hayward metric and Hawking's evaporation formalism, subject to two non-trivial additional assumptions: (i) the regular-core geometry remains stable throughout the evaporation process, and (ii) evaporation is quasi-static so that the extremal condition is reached adiabatically. Neither assumption has been verified dynamically.

Code: `hayward_remnant.py`

5. Falsifiability Table

The following table summarizes the minimum observational thresholds required to exclude — or detect — a negative-pressure core. **Entries marked with ✓ are constrained by data that already exists.**

Observable	Current Constraint	Required Measurement	Detector	Timeline	Excludes
QNM shift (Hayward)	$L/M < 0.316$ (GW150914)	$\Delta f/f \sim 10^{-4}$	ET/CE (100-event stack)	2030s	$L/M \gtrsim 0.045$
QNM shift (Hayward)	$L/M < 0.316$ (GW150914)	$\Delta f/f \sim 10^{-3}$	ET/CE (single event)	2030s	$L/M \gtrsim 0.14$
QNM shift (Bardeen)	None	$\Delta f/f \sim 10^{-4}$	ET/CE (100-event)	2030s	$e/M \gtrsim 0.027$
QNM shift (Dymnikova)	Unobservable	N/A	N/A	—	Exponential suppression
Tidal deformability ✓	$70 \lesssim \Lambda(1.4) \lesssim 580$ (GW170817)	Λ to $\sim 1\%$	LIGO A+ / ET	Now / Late 2020s	$\rho_0 \gtrsim 3 \times 10^{13}$ g/cm ³
Post-merger f_2	None	f_2 to $\sim 10\%$	ET	2030s	$\rho_0 \gtrsim 10^{12}$ g/cm ³ (stiffer EOS than SLy required)
M-R (NICER) ✓	$M = 2.08,$ $R \approx 12.4$ km	Better R precision	NICER / STROBE-X	Now	$\rho_0 \gtrsim 3 \times 10^{10}$ g/cm ³ (1σ); $\gtrsim 3 \times 10^{11}$ g/cm ³ (2σ)
Echoes	Ruled out unless horizonless	Echo SNR threshold	LIGO A+	Now	Horizonless objects
Remnant DM	$M_{\text{rem}} \lesssim 10^{18}$ g (microlensing)	Sub-lunar mass lensing	LSST / Roman	Late 2020s	$L \gtrsim 10^{-14}$ m

Key observation: The tidal deformability channel is the only one that constrains the model with data that already exists. The ringdown channel is the most theoretically robust but requires next-generation detectors. The echo channel turns out to be a false lead — echoes test for

horizon absence, not core regularity. The causality bound ($c_s^2 \leq 1$, 0/394 violations; see §3.5) is a theoretical consistency check, not an observational constraint, and is therefore not listed here.

6. Open Problems

1. **EOS degeneracy.** The TOV bound ranges from $\rho_0 \lesssim 3 \times 10^{10} \text{ g/cm}^3$ (1σ PSR mass) to $3 \times 10^{11} \text{ g/cm}^3$ (2σ), and is SLy-conditional throughout. Marginalization over EOS priors (APR4, MS1b, H4) is required for a robust constraint.
 1. **Cross-metric completeness and isospectrality.** Dymnikova QNM shifts are confirmed to be ≈ 0 (exponential suppression), but the echo-extremality coincidence should be verified numerically for Bardeen and Dymnikova. The axial–polar isospectrality splitting of 0.11% is below the leading-order WKB truncation floor; third-order Konoplya WKB or an exact numerical computation is needed to determine whether isospectrality is genuinely preserved at that level.
 1. **Full Kerr.** The slow-rotation WKB expansion is invalid at $a/M \sim 0.67$: the Lense-Thirring coupling is $\mathcal{O}(1)$ relative to the static potential peak, causing the first-order code to give unphysical spin-split frequencies (factor of $\sim 1.7\times$ from exact Kerr values). Teukolsky-equation solutions on the Hayward background are required for any spin-dependent result.
 1. **Core formation.** No dynamical mechanism is provided. Gravitational collapse producing a regular core rather than a singularity remains an open problem.
 1. **Stability.** The causality check is necessary but not sufficient. The Hayward metric's stability under non-spherical perturbations is unanalyzed.
 1. **L remains a free parameter.** The analysis constrains L from data but does not predict it. Connecting L to a fundamental scale (Planck, QCD, electroweak) requires a microphysical theory that this paper does not provide.
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References

1. J. M. Bardeen, Non-singular general-relativistic gravitational collapse, in *Proceedings of GR5*, Tbilisi, 1968, p. 174.
2. I. Dymnikova, Vacuum nonsingular black hole, *General Relativity and Gravitation* **24**, 235 (1992).
3. S. A. Hayward, Formation and evaporation of nonsingular black holes, *Physical Review Letters* **96**, 031103 (2006).
4. S. Iyer and C. M. Will, Black-hole normal modes: A WKB approach, *Physical Review D* **35**, 3621 (1987).
5. B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), Observation of gravitational waves from a binary black hole merger, *Physical Review Letters* **116**, 061102 (2016).
6. J. S. Read et al., Constraints on a phenomenologically parameterized neutron-star equation of state, *Physical Review D* **79**, 124032 (2009).
7. H. T. Cromartie et al., Relativistic Shapiro delay measurements of an extremely massive millisecond pulsar, *Nature Astronomy* **4**, 72 (2020).
8. T. Hinderer, Tidal Love numbers of neutron stars, *Astrophysical Journal* **677**, 1216 (2008).
9. B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170817: Measurements of neutron star radii and equation of state, *Physical Review Letters* **121**, 161101 (2018).
10. S. Chandrasekhar, *The Mathematical Theory of Black Holes*, Oxford University Press (1983), Ch. 4, §42–43.
11. K. D. Kokkotas and B. G. Schmidt, Quasi-normal modes of stars and black holes, *Living Reviews in Relativity* **2**, 2 (1999), §3.3.
12. S. Bernuzzi, T. Dietrich, W. Kastaun, and A. Stellwagen, Modeling the complete gravitational wave spectrum of neutron star mergers, *Physical Review D* **91**, 044056 (2015); A. Bauswein and H.-T. Janka, Measuring neutron-star properties via gravitational waves from binary mergers, *Physical Review Letters* **108**, 011101 (2012).
13. E. Berti, V. Cardoso, and A. O. Starinets, Quasinormal modes of black holes and black branes, *Classical and Quantum Gravity* **26**, 163001 (2009). [BCStab — Kerr QNM numerical tables]
14. B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170817: Constraining the nuclear matter equation of state from the neutron star tidal deformability, *Physical Review D* **97**, 021501(R) (2018). [Two-sided Λ bounds]
15. B. Toshmatov, Z. Stuchlík, and B. Ahmedov, Quasinormal modes of test fields around regular black holes, *Physical Review D* **95**, 084037 (2017).
16. A. Flachi and J. P. S. Lemos, Quasinormal modes of regular black holes, *Physical Review D* **87**, 024034 (2013).
17. Z. Li and C. Bambi, Measuring the Bardeen regular black hole with astrophysical observations, *Physical Review D* **87**, 104022 (2013).

Code repository: `tov_tidal.py`, `kerr_qnm.py`, `causality_sweep.py`, `hayward_remnant.py`,
`post_merger_f2.py`, `tidal_gw170817.py`, `conceptual_diagram.py`